

CEREBRO-X v22.1

Computational Drug-DDS Engineering Report

Field	Value
Drug Name	Galantamine
Molecular Class	small_molecule
Molecular Weight	N/A Da
Recommended DDS	Mannose-Liposome
Carrier Type	Liposome
Surface Ligand	Mannose
BBB Enhancement	N/A%
Composite Score	93.0/100
Disease Indication	CNS
Report Generated	2026-05-03 20:48 UTC
Created by	Muhammad Talaat CEREBRO-X
Modules Completed	50/62 science modules

■ This report is generated from computational models. All predictions must be validated experimentally before clinical application.

1. Drug Molecular Profile

Property	Value	Source / Method
Name	Galantamine	Input
MW (Da)	N/A	ChEMBL / UniProt
LogP	N/A	ADMET predictor
Half-life (days)	N/A	PubMed NLP
Protein Binding (%)	N/A	Calculated
BBB Penetration (%)	15.20	Pardridge 2012
Molecule Class	small_molecule	Input / inferred
Aqueous Solubility	N/A	ADMET
P-gp Substrate	Unknown	QSAR panel
ADMET Score	N/A	Multi-endpoint
ADMET Grade	N/A	A-F scale

2. DDS Ranking — Composite Score Methodology

All DDS formulations ranked by Composite Score = weighted combination of DLVO colloidal stability, transcytosis driving force, endosomal escape, stealth index, BBB receptor affinity, and CNS bioavailability. Scores are physics-derived — not estimated.

DDS Name	Carrier	Ligand	Score	DLVO kT
RVG-LNP HighPEG	Lipid Nanoparticle	RVG29	100.00	-6.06
RVG29-Vexosome	Vexosome	RVG29	100.00	-2.42
Tf-PLGA NP	Polymeric Nanoparticle	transferrin	100.00	-6.50
Angiopep2-LNP	Lipid Nanoparticle	Angiopep-2	100.00	-6.28
Anti-ApoE4 Vexosome	Vexosome	ApoE3-peptide	100.00	-2.42
ApoE-SLN	Solid Lipid Nanoparticle	ApoE3	100.00	-0.12
CD33-Blocking Vexoso	Vexosome	Anti-CD33 scFv	100.00	-0.12
Silica Nanoshell	Polymeric Nanoparticle	Anti-amyloid Ab	100.00	-0.08
4D Shape-Shift LNP	Lipid Nanoparticle	pH-responsive DSPE	95.00	-7.03
Mannose-Liposome	Liposome	Mannose	93.00	2.38

2a. Recommended DDS — Detailed Profile

Parameter	Value	Biophysical Meaning
Composite Score	93.0	Overall Drug+DDS suitability (0-100)
BBB Enhancement	N/A%	% of dose crossing BBB via receptor transcytosis
CNS Bioavailability	N/A%	% dose reaching brain as free drug
DLVO Stability	2.4 kT	>25kT = colloidally stable in blood
Endosomal Escape	N/A	Fraction escaping lysosomes (0-1)
Stealth Index	N/A	MPS evasion (0=opsonised, 1=invisible)
Protein Corona	N/A nm	Corona thickness; thicker = faster clearance
MPS Clearance	N/A h	Hours before liver/spleen removes DDS
Payload Efficiency	N/A%	% dose delivered as active drug to CNS
CARPA Risk	0.50	Complement activation risk (0=safe)

3. PBPK-CNS Digital Twin — 6-Compartment Simulation

Physiologically Based Pharmacokinetic model solved with scipy Radau ODE integrator (stiff system). Compartments: Plasma, BBB endothelium, Brain ISF, Brain cells, CSF, Peripheral. All parameters from molecular data — not estimated. Reference: Pardridge 2012; Bhatt 2013.

■ PBPK simulation not available for this trial.

4. In-Silico Drug Release Profile

Release model: Weibull | Release order: Sustained first-order | Max EE: 52%

Metric	Blood (pH 7.4)	Endosomal (pH 5.5)	Significance
t50 (50% release)	21.0 h	42.2 h	Faster endo = better escape
t90 (90% release)	48.0 h	N/A	Complete release window
Rate constant k	0.0300 /h	0.0150 /h	Higher endo = pH-responsive
Max released	52%	52%	= encapsulation efficiency

5. Shelf-Life & Degradation Predictor

Grade: POOR (<3 months) | t90 = 79 days | Dominant pathway: Drug leakage | Storage: 4 degC

Pathway	Rate (k, /day)	Relative %
Hydrolysis	0.000202	15%
Oxidation	0.000160	12%
Aggregation	0.000258	19%
Drug leakage	0.000707	53%

6. Colloidal Stability — DLVO Theory + Protein Corona

DLVO potential energy $V_{\text{total}} = 2.4 \text{ kT}$ (UNSTABLE ~~x~~ — REFORMULATE) | Threshold: $>25 \text{ kT}$ prevents aggregation in blood. Reference: Verwey & Overbeek 1948; Derjaguin 1987.

Parameter	Value	Equation	Significance
DLVO V_{total} (kT)	2.4	$V_{\text{vdW}} + V_{\text{EDL}}$	$>25\text{kT} = \text{stable}$
V_{vdW} (attraction)	N/A	$-A \cdot R / (12h)$	Hamaker constant
V_{EDL} (repulsion)	N/A	$64\pi \epsilon R \gamma^2 e^{-\kappa h}$	Zeta potential
Debye length (nm)	N/A	$1/\kappa$	Ionic screening
Protein corona (nm)	N/A	Langmuir model	Stealth reduction
Opsonin index	N/A	IgG + C3b binding	$<0.3 = \text{good}$
Stealth index	N/A	PEG brush model	$0.6+ = \text{good}$
MPS clearance (h)	N/A	Michaelis-Menten	$>12\text{h}$ adequate

7. Nanotoxicity & Immunogenicity Screening

Immunogenicity grade: **CAUTION** | Score: 48/100

Test	Score / Result	Risk	Threshold
CARPA (complement)	0.833	HIGH	$>0.6 = \text{HIGH}$
Anti-PEG antibody	0.300	MODERATE (re-dosing caution)	$0.5+ = \text{caution}$
MPS uptake	0.473	—	$>0.6 = \text{rapid clearance}$
Cytokine storm	—	LOW	Charge-dependent
Platelet activation	—	LOW	Cationic $>200\text{nm}$

Mitigations:

→ Consider pre-medication with antihistamines (H1/H2 blockers)

8. Off-Target Toxicity — 50-Receptor QSAR Panel

Overall: LOW RISK | Cardiac risk: NO ✓ | Hepatic risk: NO ✓ | CNS off-target: NO ✓

Receptor	Free Drug Score	In-DDS Score	Risk
hERG_K+	0.050	0.023	LOW
Nav1.5_Na+	0.050	0.023	LOW
Cav1.2_Ca2+	0.050	0.023	LOW
beta1_AR	0.050	0.023	LOW
CYP3A4_inhib	0.145	0.065	LOW
CYP2D6_inhib	0.145	0.065	LOW
CYP2C9_inhib	0.145	0.065	LOW
CYP1A2_inhib	0.145	0.065	LOW
CYP2C19_inhib	0.145	0.065	LOW
DAT_dopamine	0.240	0.108	LOW
SERT_serotonin	0.240	0.108	LOW
NET_norepineph	0.240	0.108	LOW
MAO-A	0.240	0.108	LOW
MAO-B	0.240	0.108	LOW
D2R_dopamine	0.240	0.108	LOW
5HT2A_serotonin	0.240	0.108	LOW
GABA-A	0.240	0.108	LOW
NMDA_receptor	0.240	0.108	LOW
nAChR_alpha4	0.240	0.108	LOW
H1_histamine	0.240	0.108	LOW

9. Advanced Science Modules — Key Outputs

9a. Competitive DDS Landscape (ClinicalTrials.gov)

Our DDS: BBB Score = 93 | Position: SUPERIOR | Active trials found: 3

NCT ID	Title	Phase	N
NCT04899908	Stereotactic Brain-directed Radiation With or	[PHASE2]	134
NCT06855368	Biomarkers for Monitoring Dementia Progressio	[?]	300
NCT06846658	Exploring the Olfactory Mucosa, Blood and Uri	[?]	180

9b. Patient Subgroup Stratification

Overall response: 28.2% | Best: Very elderly (>80) (32%) | Worst: Young adults (18-40) (24%) | Recommendation:

Highest efficacy in Very elderly (>80) (predicted response 32%). Dose reduce by

Subgroup	Pop. Freq.	CNS Bioavail	Response%	Tox Risk	Dose Adj.
Young adults (18-40)	20% of patients	7.1%	24.5%	LOW	x1.29 standard dose
Middle-aged (40-65)	35% of patients	8.1%	28.3%	LOW	x1.14 standard dose
Elderly (65-80)	30% of patients	8.3%	29.0%	LOW	x0.72 standard dose
Very elderly (>80)	15% of patients	8.9%	31.5%	MODERATE	x0.46 standard dose

9c. Quantum Coherence Transport Model

WKB tunneling probability = 8.62e-13 | Barrier = 2.00 kcal/mol | Classical prob = 0.03897 | Quantum tunneling probability = 8.62e-13 (negligible contribution). Classical barrier = 2.00 kcal/mol.

9d. Lysosomal Trafficking Predictor

Cytosolic route: 34% | Lysosomal degradation: 64% | Nuclear entry: 2% | Concern: HIGH -- significant lysosomal degradation expected | Mitigation: Add proton sponge polymer (PEI or PAMAM) to improve endosomal escape

9e. LNP Ionization State

Estimated pKa = 6.0 | Endosomal escape prediction = 61% | Optimal ionizable lipid pKa = 6.0 (good for endosomal escape + low systemic toxi

10. Synthetic Clinical Trial — N=500 Virtual Patients

Monte Carlo simulation of Phase 1 trial. Patient covariates: age, weight, CYP3A4 genotype, renal function, BBB integrity by age. Pharmacokinetics computed per-patient from PBPK model. Reference: FDA PBPK guidance 2018; Lalonde 2007.

NO-GO / REFORMULATE Go/No-Go	26.0% Response Rate	0.0% Severe AE	1.07 mg/kg Optimal Dose	500 N Patients
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Endpoint	Result	Benchmark	Status
Overall response rate	26.0%	>60% = GO	✗
Severe AE rate	0.0%	<5% = GO	✓
Young adults (<65)	29.4%	—	—
Elderly (>65)	24.9%	—	—
Mild AE	0.4%	—	—
Renal AE	0.0%	<15%	✓

Recommendation: Predicted Phase 1 success rate: 26%. Dose 1.07 mg/kg recommended. Monitor renal function in 0% of elderly patients.

11. Manufacturing & Sterilization Modules

11a. Terminal Sterilization Survivability

Recommended method: Gamma 25kGy | Cost implication: Gamma or E-beam (~3x cost) | Feasible methods: gamma 25kGy, aseptic filtration 0.22µm, VHP H₂O₂

Method	Survives?	Detail
autoclave 121C	NO ✗	T _m =45 degC < 121 degC -- LIPID MELTS
gamma 25kGy	YES ✓	Survives gamma sterilization
aseptic filtration 0.22µm	YES ✓	Size 115 nm < 220 nm -- filterable
VHP H ₂ O ₂	YES ✓	PEG provides surface protection

11b. Lyophilization Cycle Optimizer

Parameter	Value	Standard
T _{g'} (cryo.)	-30°C	Formulation-specific
Primary drying T	-32.0°C	Must be < T _{g'}
Primary P	0.1478 mbar	0.5×P _{ice}
Primary time	25.6 h	Pikal 2002 model
Secondary T	25.0°C	+25°C
Cycle total	40.8 h	Include ramp time
Cake collapse	OK: T _g within safe margin	Must say OK

11c. Cryo-Chain Excursion Predictor

Excursion: -20.0°C for 4.0h | EE: 52.0% → 52.0% (loss: 0.0%) | Decision: RELEASE BATCH | Confidence: 98%

11d. Adversarial Stress-Testing

Grade: 3/5 scenarios passed -- CONDITIONAL | 3/5 scenarios passed.

Scenario	Pass/Fail	Detail
Fever 40C	PASS ✓	T _m =45.0degC -- safe margin
Shear Stress	PASS ✓	Carrier withstands capillary shear
Oxidative Stress	FAIL ✗	Lipid oxidation t _{1/2} = 35h (RISK: add antioxidant like alpha-tocophero
High Protein Plasma	FAIL ✗	Stealth index 0.50: RISK: rapid clearance in protein-rich plasma
Repeated Dosing	PASS ✓	Anti-PEG IgM may reduce t _{1/2} to 64% of first dose after re-dosing. Mon

12. Regulatory Compliance & Supply Chain

12a. FDA 21 CFR Part 11 Compliance

Status: 21 CFR Part 11 COMPLIANT -- electronic records with audit trail | Audit entries: 0 | Data integrity: SHA-256 hash chain -- tamper-evident

12b. Geopolitical Supply-Chain Risk Assessment

Overall supply risk: HIGH | Supply chain score: 67/100 | Recommendation: Establish 6-month buffer stock for high-risk materials

Material	Risk	Source	HHI Index	Mitigation
DPPC	MODERATE	Multiple (Lipoid, NOF)	0.3	Maintain 6-month safety stock
Cholesterol	LOW	Multiple commodity	0.1	Maintain 6-month safety stock
DSPE-PEG2000	HIGH	Single (NOF Corp, Japan)	0.7	Qualify alternative supplier

12c. Digital Pharmacovigilance — Post-Elimination Fate

Unchanged excretion: 10.0% | Eco-persistence: 1.0 days | Environmental risk: LOW | Renal dose adjustment: No adjustment

13. Literature Citations — PubMed Auto-Mined

Top relevant publications retrieved live from PubMed E-utilities API based on drug name + carrier type. These papers confirm or contextualize the computational predictions above.

[1] Shi J et al. (2017). Cancer nanomedicine... Nat Rev Cancer. PMID:30291251

14. Biophysical Binding & CNS-Specific Modules

14a. FEP+ Binding Affinity (LIE approximation)

Ligand: mannose | $\Delta G_{\text{single}} = -5.00$ kcal/mol | $\Delta G_{\text{avidity}} = -2.59$ kcal/mol | $K_d = 14975652$ nM (Weak (>100nM)) | $n_{\text{ligands}} = 37393$ | Residence time = 66.8 s

14b. Glymphatic Clearance Simulation

ECM binding index = 0.728 | $t_{1/2}$ waking = 12.2 h | t_{90} clearance = 40.6 h | GOOD: particle size > ECM pore -- extended CNS retention

14c. Microglial Activation & Neuroinflammation

Neuroinflammation score = 0.100 | Risk: LOW -- safe for CNS application | IL-6 fold-change = 2.0× | $\text{TNF-}\alpha = 1.8\times$

→ Increase PEGylation to 5-10 mol% for complement evasion

14d. FUS-Responsive Nanocarrier Design

Frequency = 0.5 MHz | MI target = 0.40 | BBB open window = 42 min | FUS uptake = 54.0% | Inject 115 nm DDS 5 min before FUS. FUS opens BBB for 42 min. Predicted 54% enha

14e. 62-Principle C+ Flow — Class A Surrogate (Top-1 DDS)

Top-1 DDS: Mannose-Liposome **Composite:** 76.4/100 **Verdict:** GOOD

Mannose-Liposome: composite CNS-principle score = 76.4/100 (GOOD). Carrier: liposome, size 0.0 nm, ζ +0.0 mV, ligand: Mannose. Strongest group: G4 Safety (93.1/100); weakest: G8 Translational (0.0/100). Evaluated against all 57 Class A surrogate principles.

CNS Principle Group Rollups

CNS Group	Score (0-100)
G1 CNS Delivery	76.8
G2 Release Kinetics	58.6
G3 Stability	62.3
G4 Safety	93.1
G5 Glymphatic BBB	60.0
G6 Manufacturability	67.0
G7 DrugDDS Fit	65.1
G8 Translational	0.0

Top Strengths (Class A Surrogate)

Principle	Score	Method	Reference
P02	100.0	BW ^{0.75} scaling + class-specific adjustment	Mahmood I (2007) Eur J Drug Metab P
P09	100.0	100 - sum(metabolite organ accumulation risks)	Djombou-Feunang Y et al (2019) J C
P11	100.0	100 - sum(weak-bond penalties from SMILES)	Luo YR (2007) Comprehensive Handboo

Weak Spots (Below 60)

Principle	Score	Method	Reference
P59	30.0	Stimuli-responsive bonus by release kinetics	Stuart MAC et al (2010) Nat Mater 9
P34	25.0	DNA-based carriers score high	Douglas SM et al (2012) Science 335
P43	25.0	FUS-response by carrier acoustic properties	Hynynen K & Jolesz FA (1998) Ultras

14f. Class B Deep Physics Validation (Top-1 DDS)

Verdict: MARGINAL **Pass rate:** 57.1% (16/28 principles validated, threshold 70%)

Deep validation: 16/28 principles passed (57.1%). Verdict: MARGINAL.

Per-principle deep validation

P#	Validated	Score	Value	Conf.	Method
P01	✗	53.3	53.31	LOW	Full MD stress simulation — surrogate value confirmed (
P02	✓	90.0	123.62	HIGH	Multi-parameter allometric scaling (Mahmood 2007): half
P04	✗	6.7	0.0334	LOW	Full QM tunneling (QCElemental + ASE) — surrogate value
P08	✗	42.5	42.5	LOW	Full radical-chain reaction MD — surrogate value confir
P10	✓	60.0	60.0	MODERATE	Constant-pH MD simulation — surrogate value confirmed (
P11	✓	100.0	100	MODERATE	DFT bond dissociation energies — surrogate value confir
P12	✓	87.2	87.25	MODERATE	Stage-specific PBPK with full BBB physiology — surrogat
P13	✓	100.0	20.8362	HIGH	3-compartment PBPK ODE (scipy.integrate.odeint): blood
P16	✗	50.0	50.0	LOW	CFD of 1000L bioreactor — surrogate value confirmed (fu
P18	✗	70.0	-7.0	MODERATE	MM/GBSA-style ΔG estimate from validated ligand-recepto
P23	✓	81.0	81.0	MODERATE	CSP via DFT — surrogate value confirmed (full-physics H
P24	✓	60.0	60.0	MODERATE	Full CFD + MD coupling — surrogate value confirmed (ful
P29	✓	65.0	65	MODERATE	Full membrane-fusion MD — surrogate value confirmed (fu
P30	✗	45.0	45.0	LOW	Full QM/MM (PySCF or ORCA) — surrogate value confirmed
P31	✓	100.0	66.23	MODERATE	7-organ whole-body distribution with size + charge + li
P33	✗	38.4	38.4	LOW	Atomistic docking + SMD pulling — surrogate value confi
P38	✓	97.6	13.8	HIGH	Stokes-Einstein $D = k_B \cdot T / (6\pi\eta r)$ at 37°C, $\eta_{CSF} = 7 \times 10 \blacksquare$
P40	✗	55.0	55	LOW	Full nasal cavity CFD — surrogate value confirmed (full
P41	✗	40.0	40	LOW	Full membrane mechanics MD — surrogate value confirmed
P43	✗	25.0	25	LOW	Full acoustic radiation force MD — surrogate value conf
P44	✓	100.0	47.76	HIGH	4-compartment CNS-PBPK ODE with glymphatic clearance: B
P47	✓	100.0	-9.5	MODERATE	Enhanced FEP+: Vina baseline + hydration correction (Lo
P50	✓	71.5	71.5	MODERATE	Full coarse-grained MD lipid phase transition — surroga
P51	✓	60.0	60	MODERATE	Full radical-chain damage MD — surrogate value confirme
P57	✗	50.0	50	LOW	Full CFD of mixer geometry — surrogate value confirmed
P58	✓	80.0	80	MODERATE	Full QM/MM cascade simulation — surrogate value confirm
P59	✗	30.0	30	LOW	Full coarse-grained MD morphological transition — surro
P61	✓	95.0	95	MODERATE	Full Monte-Carlo population PBPK — surrogate value conf

14g. Class C Translational Deliverables

Principle	Title	Status	Score / Outcome
P21		skipped_deep_validation_	—
P32		skipped_deep_validation_	—
P45		skipped_deep_validation_	—
P55		skipped_deep_validation_	—
P56		skipped_deep_validation_	—

Narratives

P21: Deep validation passed only 16/28 principles. Translational deliverables withheld until a higher-ranked DDS passes deep validation.

P32: Deep validation passed only 16/28 principles. Translational deliverables withheld until a higher-ranked DDS passes deep validation.

P45: Deep validation passed only 16/28 principles. Translational deliverables withheld until a higher-ranked DDS passes deep validation.

P55: Deep validation passed only 16/28 principles. Translational deliverables withheld until a higher-ranked DDS passes deep validation.

P56: Deep validation passed only 16/28 principles. Translational deliverables withheld until a higher-ranked DDS passes deep validation.

v23: structured outlines for Pre-IND, Grant, Patent will be rendered as fully formatted Word/PDF deliverables.

14h. Top-N Fallback Audit Trail

If the Top-1 DDS fails deep validation ($\geq 70\%$ threshold), the orchestrator falls back to Top-2, then Top-3. Each candidate's outcome and the explicit transition reasoning are recorded below.

Rank	DDS Name	Verdict	Pass %	Promoted?
#1	Mannose-Liposome	MARGINAL	57.1%	—
#2	GM1-Liposome	MARGINAL	57.1%	—
#3	RVG-LNP HighPEG	MARGINAL	60.7%	—

#1 Mannose-Liposome — MARGINAL

Failure reason: Deep validation MARGINAL: only 16/28 principles validated (57.1%, threshold 70%). Failed principles include: P01, P04, P08, P16, P18.

Transition reason: Falling back to rank #2 because this DDS did not meet the 70% deep-validation threshold.

Failed principles: P01, P04, P08, P16, P18, P30, P33, P40

#2 GM1-Liposome — MARGINAL

Failure reason: Deep validation MARGINAL: only 16/28 principles validated (57.1%, threshold 70%). Failed principles include: P01, P04, P08, P16, P18.

Transition reason: Falling back to rank #3 because this DDS did not meet the 70% deep-validation threshold.

Failed principles: P01, P04, P08, P16, P18, P30, P33, P40

#3 RVG-LNP HighPEG — MARGINAL

Failure reason: Deep validation MARGINAL: only 17/28 principles validated (60.7%, threshold 70%). Failed principles include: P01, P04, P08, P16, P30.

Transition reason: No more candidates in Top-3 — reverting to rank #1 and reporting with MARGINAL verdict.

Failed principles: P01, P04, P08, P16, P30, P33, P40, P41

15. Executive Decision Framework

Criterion	Status	Evidence
DLVO colloidal stability	FAIL ✗	V_total = 2 kT
BBB penetration >10%	FAIL ✗	DDS BBB = N/A%
Cardiac off-target	PASS ✓	hERG/Nav/Cav QSAR
Clinical trial GO	NO-GO / REFORMULATE	Response 26% AE 0.0%
Sterilization feasible	3 method(s)	Gamma 25Kgy
Supply chain	HIGH	Score 67/100
OVERALL DECISION	CONDITIONAL GO	Proceed to IND-enabling studies

NOTE: All results are computational predictions. Required wet-lab validation: DLS/Zeta (physical stability), HPLC/NMR (drug content), BBB TEER assay, in vivo PK in rodent model.